

Beginner Math Roadmap

How I Would Relearn Math If I Could Start Over

Introduction

If I had to relearn math from the very beginning, I would not waste time wandering between random books, YouTube channels, or problem sets. Instead, I would follow a structured progression: master the basics, then gradually increase rigor while keeping problem-solving at the center.

The biggest mistake people make is **jumping too quickly** into advanced material without first building comfort with fundamentals or learning how to write proofs. This roadmap avoids that mistake.

1. Foundations (Algebra - Pre-Calculus)

Resources:

- *Khan Academy – Algebra, Geometry, Trigonometry, Pre-Calculus* (all free)

Why:

This stage builds the muscle memory. You'll drill algebraic manipulation, geometric reasoning, trigonometric identities, and functions. Pre-calculus introduces the language of limits and functions that will appear everywhere later.

How to Study:

- Set a daily goal: 30–60 minutes of practice.
- Don't move on until you can do problems *without hints*. Only move to the next section when you passed the final exam of a course without any errors. You will most likely have to retry many times, but this is where the learning happens.
- Keep a notebook where you write out solutions longhand. This will help a lot with problem solving, because your brain remembers problem solving techniques even better if you write out the complete solution.

2. Calculus I

Primary Book: *Calculus by Michael Spivak*

Alternative (if you want a softer intro): *Calculus Made Easy* by Silvanus Thompson

Why Spivak:

- It teaches calculus *from first principles*, building rigor and intuition together.
- It introduces proofs alongside computation, so you learn “why” as well as “how.”

How to Study:

- Read each section actively, pencil in hand. Write down the most important definitions and theorems. Have them easily available and not lost in notes.
- After examples, cover the solution and try to redo it yourself.
- End each week by summarizing key theorems in your own words. Theorems like the mean value theorem can be hard to understand at first, but if you draw pictures and actually spend some time reading the theorem and thinking about it, it will make sense in the end.

3. Proofs & Logic (in Parallel with Calculus II)

Book: *Book of Proof* by Richard Hammack (free PDF)

Why Now:

By the time you start Calculus II, you’ll see sequences, series, and deeper concepts that require proof techniques. Hammack’s book is perfect for beginners: it introduces logic, sets, functions, induction, and proof writing clearly. It’s actually the single book that made me good at math and proofs, and after reading this book, I found other proof based topics much easier to work through.

How to Study:

- Do *every* exercise, not just read passively.
- Try to prove results before looking at solutions. If you copy proofs from the internet or solution manual, you don’t develop your brain into being able to solve problems and construct proofs.

- Discuss proofs out loud, if you can explain it verbally, you understand it. Even if you are studying alone, just talking out loud to yourself explaining the proof can help a lot.
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4. Multivariable Calculus + Linear Algebra (in Parallel)

Linear Algebra Resource: *Introduction to Linear Algebra* by Gilbert Strang

Multivariable Calculus Resource: *Multivariable Calculus* by Peter Lax

Why in Parallel:

- Multivariable calculus teaches you to think in higher dimensions.
- Linear algebra gives the language (vectors, matrices, transformations) that underlies almost everything in advanced math and applications like machine learning, physics, and finance.
- These topics, especially linear algebra will come up later in almost every single math course you take later. They will also come up in applications like machine learning etc. It's very important to spend some time on these topics and actually understand them.

How to Study:

- For Strang: focus on intuition - visualizing vector spaces, orthogonality, projections. Draw out the spaces, and the maps between them. Focus on actually visualizing the relationships between the null space, row space, column space and the left null space.
- For Lax: expect more rigor. Push through it, even if it feels heavy.
- Supplement with MIT OpenCourseWare (Strang's video lectures are legendary).
- After finishing Strang's *Introduction to Linear Algebra*, a great next step is Axler's *Linear Algebra Done Right* if you want a deeper and more abstract challenge. When I first tried reading Axler a couple of years ago, I struggled a lot — not because the material was impossible, but because I lacked a solid foundation in proofs. That's why books like Hammack's *Book of Proof* are so important: they give you the logical tools needed to fully appreciate a text like Axler.

5. Where to Go Next (Optional, After the Core)

Once you complete the above sequence, you'll be at the level of a strong university sophomore. From here, paths branch depending on goals:

- **Applied Math / Quant:** Differential equations, probability, statistics, numerical methods.
- **Pure Math:** Real analysis (start with *Understanding Analysis* by Abbott), abstract algebra, topology.
- **Computer Science / ML:** Optimization, linear algebra applications, probability theory, discrete math.

Key Study Habits

1. **Don't just read - solve.** Math is not passive. Aim for >70% of your time on exercises.
2. **Space repetition.** Revisit problems you got wrong a week later.
3. **Mix easy + hard.** Build confidence with basics, then stretch with challenge problems.
4. **Explain to someone else.** If you can teach it, you've mastered it.
5. **Consistency over intensity.** One hour a day beats cramming 7 hours on Sunday.

Final Thoughts

If I could start math over, I'd follow this path exactly:

1. Khan Academy for fundamentals.
2. Spivak for Calculus I.
3. Hammack for proofs in parallel with Calculus II.

4. Strang (Linear Algebra) + Lax (Multivariable Calculus).

This path balances intuition, rigor, and problem-solving. It avoids the trap of being “good at calculations” but bad at understanding. If you stick with it, you’ll have a foundation strong enough for quant interviews, graduate-level courses, and advanced research.

